Unsupervised Learning Report Rubric
CS7641: Machine Learning
Fall 2026

Schedule

- **11/9/2026 (Monday):** UL Grading Begins & Overview of Rubric
- **11/11/2026 (Wednesday):** UL First 2 Calibration
- **11/13/2026 (Friday):** UL Calibration with Examples and Head TAs Check
- **11/17/2026 (Tuesday):** UL 33–50% Completed
- **11/20/2026 (Friday):** UL 50–80% Completed
- **11/24/2026 (Tuesday):** UL 100% Completed; **early grade-posting target** if grading and final checks are complete
- **11/25/2026 (Wednesday):** UL Posted (fallback if not posted Tuesday)
- **11/26/2026 (Thursday):** Thanksgiving Day (holiday; no grading planned)

Thanksgiving planning: Thanksgiving 2026 falls on Thursday, November 26. If we can finish grading and verification early, we can post grades on **Tuesday, November 24** rather than Wednesday, and avoid needing to grade over the Thanksgiving holiday. Please prioritize completing and checking your assigned reports ahead of that date when feasible.

Total: 100 points + 5 extra credit

This rubric is aligned to the finalized Fall 2026 Unsupervised Learning (UL) Report description and FAQ. The five-part Fall assignment restores the cluster-derived neural-network experiment, so the scoring distribution differs from Summer 2026.

How to Use This Rubric

This rubric supports **consistent grading across TAs** while rewarding controlled, evidence-backed scientific analysis.

Each line item has a **fixed point value**. Award either all points for that item or zero. Do not award partial credit inside an item unless it is explicitly divided into smaller scored items. Award points only when the report demonstrates the requirement clearly and correctly.

The main report must stand alone. REPRO may contain full hyperparameter sweeps, extra figures, seeds, code and environment details, and extended diagnostics. Do not require every underlying run in the eight-page report, but do require the main result and interpretation there.

Avoiding Double Jeopardy

Do not deduct twice for the same error unless it independently prevents two distinct requirements from being demonstrated. For example, missing ICA does not automatically remove all Discussion points if the student otherwise provides a meaningful synthesis. Missing experiment data should remove the relevant experiment points; broader interpretation points should be judged on the evidence that remains.

TA pre-grading checks are for access holds, required-task failures, grading boundaries, and escalation. They are not an extra set of point deductions. An unverifiable required submission artifact can trigger a **temporary overall zero**, but staff should record the provisional earned score, update the applicable access criterion after verification, and restore the earned score. The access hold is not an additional permanent point deduction.

Minimum Standard for Method Exploration

When an item calls for multiple settings, expect normally **at least three meaningful values** of a central scalar choice (for example k, GMM components, or DR dimensions), unless a smaller exploration is justified by computational constraints or prior evidence. The values should span a meaningful range rather than being cosmetic variants. Do not require an exhaustive grid search.

For randomized methods, evaluate the student's seed claims according to the evidence presented. RP should include variability across multiple seeds; a claim of superior stability or performance should not rest on one selectively favorable run.

Required Task

Students must complete the task below. The UL dataset and classification context are fixed even though Parts 1–3 are unsupervised.

- **Dataset:** Diabetes Health Indicators Dataset derived from the 2015 Behavioral Risk Factor Surveillance System (BRFSS), studying General Health.
- **Required file:** `diabetes_012_health_indicators_BRFSS2015.csv`.
- **External label and NN target:** `GenHlth`, five classes: 1 = Excellent, 2 = Very Good, 3 = Good, 4 = Fair, 5 = Poor.
- **Working sample:** Approximately 20,000 instances, matching the SL/OL sample unless a change is disclosed and justified.
- **Parts 1–3:** Original-space K-Means and GMM/EM; PCA, ICA, RP; and both clusterers after each DR method.
- **Parts 4–5:** Original/PCA/ICA/RP input comparison for the established PyTorch NN, plus original/K-Means-feature/GMM-feature NN comparison.
- **Required NN evidence:** Validation loss, Accuracy, Macro-F1, balanced accuracy, confusion-matrix or comparable class-level evidence, and per-class precision/recall/F1 discussion; efficiency or runtime comparison in the NN parts.

`GenHlth` must not be included as a feature for unsupervised fitting. It may be used *after* fitting for external evaluation and in Parts 4–5 as the supervised target. The ordered response values do not change the required NN task into regression.

Algorithm Review for TAs

The purpose is not to make clusters match the five health labels. It is to examine which geometric or probabilistic structure the original features contain, how representations alter that structure, and whether DR or cluster-derived features influence the previously justified PyTorch NN.

BRFSS contains binary health/behavior indicators, ordinal responses, and numeric measurements with different scales. These characteristics affect Euclidean distances, Gaussian approximations, variance-based transformations, and downstream classification. The five health categories are imbalanced; a higher Accuracy can coincide with weaker Macro-F1, balanced accuracy, or recall for less-represented Fair/Poor categories.

Do not require a particular accuracy, ARI, silhouette score, or algorithm winner. Moderate or weak label agreement can be scientifically meaningful if students identify what clusters capture and support their interpretation. A method that hurts NN performance may still be informative when comparisons are controlled and the limitations are explained.

Part 1: K-Means and EM/GMM

K-Means searches for compact groups under Euclidean geometry; GMM models data through Gaussian mixture components and may assign soft membership. Five clusters/components is only a reference point, *not* the mandatory or optimal choice. Require defensible preprocessing and parameter selection, suitable internal diagnostics, external label-alignment evidence after fitting, and some interpretation of cluster size or feature profiles.

GMM Gaussian assumptions are approximations for the mixed binary, ordinal, and numeric BRFSS data. A lower BIC does not imply better health-category alignment. Likewise, low ARI does not by itself establish clustering failure.

Part 2: PCA, ICA, and RP

PCA preserves variance; students should examine eigenvalues/explained variance and the choice of retained dimensions. ICA estimates non-Gaussian independent components; students should consider kurtosis or related diagnostics and convergence. RP uses random projection; students should examine distance preservation or distortion and seed variability. No single diagnostic is directly interchangeable across all three algorithms.

Part 3: Clustering after DR

There are **six required combinations**. Students must explain or defend post-DR clustering settings and compare clustering with the original feature space. Be cautious about conclusions based on raw inertia, likelihood, or BIC across spaces with different dimensions or scales; these quantities may not be directly comparable.

Part 4: NN after DR

Use the best justified PyTorch NN architecture and training recipe informed by SL and OL. Changing input-layer width for the transformed dimension is expected; a new architecture search is not. All learned preprocessing and PCA/ICA/RP transformations used for NN testing must be fit using training features only. Evaluate the original-feature NN against PCA, ICA, and RP with the same splits and comparable training conditions.

Part 5: NN with Cluster-Derived Features

This is required for Fall 2026. Students construct inputs from *original-feature-space* K-Means and GMM configurations selected in Part 1. Appropriate features include one-hot cluster assignments, distances to K-Means centroids, and GMM posterior probabilities. They may choose **replacement or augmentation**; both designs are not required.

For this supervised experiment, students **refit the selected raw-space cluster configurations on training features only** and transform validation/test without refitting. They compare the original-feature NN, K-Means-feature NN, and GMM-feature NN. Part 5 does *not* require NN experiments for all six DR + clustering combinations from Part 3.

Label Use, Leakage, and the Test Set

For exploratory Parts 1–3, clustering and DR may be fit on the working sample without `GenHlth`; labels may be overlaid afterward for interpretation. For Parts 4–5, the original SL/OL split must be preserved, and all fitted

preprocessing/DR/clustering used to create NN inputs must be trained on the training subset only. Validation labels may guide supervised NN selection, but the held-out test set must not be used to select components, cluster counts, model variants, or favorable RP seeds.

Rubric

Total Points: 100 + 5 Extra Credit

TA Start Check

Complete these before scoring. These checks do not create extra scored penalties.

☐ Both PDFs, `UL Report` and `REPRO UL`, are present and readable.

☐ The student uses `diabetes_012_health_indicators_BRFSS2015.csv`, not a binary diabetes source file.

☐ `GenHlth` is the five-class target and is excluded from unsupervised input features.

☐ The approximately 20,000-instance working sample and SL/OL splits are reused, or material changes are explained.

☐ K-Means/GMM, PCA/ICA/RP, all six DR + clustering combinations, and both NN parts are addressed.

☐ Both NN parts use a justified SL/OL PyTorch backbone and training-only learned transforms.

☐ A GT Enterprise Overleaf **Edit Link** is provided and accessible.

☐ The GT Enterprise GitHub final commit hash and repository are accessible.

☐ Run instructions, model settings, and random seeds are available in REPRO.

☐ The AI Use Statement is present.

☐ Any unusual hidden-cue echoes have been considered only as a **non-scoring AI-use follow-up flag**, never as a requirement or proof.

☐ The main report uses the IEEE conference format and is within the eight-page scoring limit.

☐ Results and Discussion are written as formal analysis rather than an unsupported bullet checklist.

Check 1: Readability

If the main report cannot be opened or evaluated, assign a zero and document the problem. Missing or inaccessible supplemental artifacts should follow the temporary-verification process rather than an automatic final zero.

Check 2: Required BRFSS General Health Task

Confirm use of the required 2015 BRFSS source file, `GenHlth` as five-class external label and supervised NN target, and appropriate unsupervised fitting without the target as input. Wrong source data, switching to `Diabetes_012` as target, binary collapsing, or treating `GenHlth` as regression are substantive required-task failures and receive a zero rather than a temporary access hold.

Check 3: Submission Access

Check that the Overleaf **Edit Link**, found in the main report, REPRO, or Canvas submission comment, opens the student's **Georgia Tech Enterprise Overleaf** project with the required access. A read-only View Link alone does not satisfy Fall 2026 requirements. Check the final commit hash in **GT Enterprise GitHub**, not a personal repository, and both required PDFs.

Action: If required access or an artifact cannot be verified, score the available work, provide substantive feedback, and enter a **temporary overall zero** pending verification. Keep the underlying criterion score provisional until access is checked. After verification, award the applicable access points and restore the earned overall score; do not apply a second penalty for the same verification issue. If access remains noncompliant, refer to the Head TA.

Check 4: Page Limit

Do not read or score content beyond page 8 of the main report, including references beyond the limit. REPRO may contain supporting technical material but cannot replace core experimental evidence and reasoning in the main report.

Check 5: Draft-Style Writing

Check whether Results, Discussion, and Conclusion contain integrated technical prose rather than repeated bullet-only summaries. Bullets are permissible for settings and short procedures. If excessive checklist writing replaces actual analysis, apply the relevant scored interpretation requirements to the missing evidence; escalate an essentially unevaluable draft to the Head TA rather than inventing a new percentage penalty.

Check 6: AI-Use Follow-up Flag (Non-scoring)

The Fall assignment source contains a deliberately unobtrusive instruction asking for a dark-blue/purple gradient in graphs and repeated use of the word `whirling`. If a submission follows either, especially the repeated unrelated word, **flag it for a discretionary AI-use/provenance follow-up** and review the student's AI Use Statement under the ordinary course process. This is a *screening signal only*, not proof of AI use or an integrity violation, a graph palette can arise independently, and AI assistance itself is allowed with disclosure.

Rubric Points

All listed criteria are all-or-nothing. Record scores under each criterion, then sum to the section totals shown below.

1 Compliance (16 points)

Data Compliance (3 points)

- **(2 pts) BRFSS working sample.**
  Award if: The student uses approximately 20,000 observations from the required BRFSS source, consistent with the SL/OL sample or with a disclosed and justified change.
  Do not award if: The sample is substantially different without explanation, the working sample is unclear, or the student substitutes another data subset without justification.

- **(1 pt) General Health task definition.**
  Award if: The report correctly identifies `GenHlth` classes 1–5 as the external evaluation label and supervised NN target, while excluding it from unsupervised predictors.
  Do not award if: The target/label mapping is missing or ambiguous, or the report does not demonstrate the required five-class context.

Submission Access (4 points)

- **(2 pts) GT Enterprise Overleaf.**
  Award if: The required **Edit Link** to a GT Enterprise Overleaf project is provided and successfully verified, including after resolution of a temporary access hold.
  Do not award if: The student has not provided a valid Enterprise Edit Link after the verification opportunity; a temporarily unverifiable link triggers a hold, not an immediate permanent deduction.

- **(2 pts) GT Enterprise GitHub.**
  Award if: The final commit hash and corresponding GT Enterprise GitHub repository are supplied and verified, including after resolution of a temporary access hold.
  Do not award if: The final hash or Enterprise repository remains missing, invalid, or noncompliant after verification; a temporary access problem alone is handled as a hold.

Reproducibility (3 points)

- **(2 pts) Linux run instructions.**
  Award if: REPRO includes the environment/package requirements, commands, data paths, split/seed information, and directions sufficient to reproduce results on a standard Linux machine.
  Do not award if: The instructions are absent, omit critical steps, or cannot reasonably reproduce the reported experiments.
- **(1 pt) Experiment specifications.**
  Award if: REPRO records relevant clustering/DR choices, NN configurations, component counts, metric procedures, and Part 5 cluster-feature construction sufficiently to support the reported exhibits.
  Do not award if: Important settings, feature construction, or metric procedures are missing or disconnected from the reported results.

Presentation (2 points)

- **(1 pt) IEEE conference format.**
  Award if: The submitted main report uses the required IEEE conference paper format.
  Do not award if: The main report uses a different document format without a permitted exception.
- **(1 pt) Readable figures and tables.**
  Award if: The central figures and tables, including axes, labels, captions, and legends, are legible at normal grading zoom.
  Do not award if: Multiple important exhibits are cropped, unreadable, poorly labeled, or otherwise cannot be evaluated.

Attribution (4 points)

- **(2 pts) Peer-reviewed sources.**
  Award if: At least two relevant peer-reviewed papers beyond course materials are cited in IEEE style and used in the report body to justify methods or interpret results.
  Do not award if: Fewer than two qualifying sources are used, papers are merely listed, or only textbooks, blogs, documentation, tutorials, or non-peer-reviewed preprints are offered.
- **(2 pts) AI Use Statement.**
  Award if: The report identifies AI tools and assistance, if any, and confirms review and understanding of assisted material; an explicit statement of no AI use is sufficient when accurate.
  Do not award if: The statement is absent, vague, or does not describe the tools/assistance and the student's verification of assisted content.

2 Study Design (10 points)

Goal: Establish a hypothesis-driven, fair, and leakage-safe experimental study that extends SL/OL.

- **(2 pts) Preprocessing rationale.**
  Award if: The report explains scaling/encoding decisions for binary, ordinal, and numeric BRFSS features and why those decisions matter for clustering and DR.
  Do not award if: Preprocessing is unexplained, described only as a default pipeline, or its effects on feature geometry are ignored.
- **(2 pts) SL/OL continuity.**
  Award if: The report identifies the reused working sample, train/validation/test partitions, and NN baseline; material differences from SL/OL are disclosed and justified.
  Do not award if: The earlier sample/splits or NN baseline cannot be identified, or material changes are unexplained.
- **(2 pts) External-label discipline.**
  Award if: `GenHlth` is excluded from unsupervised fitting and used only afterward for alignment/interpretation, or as the supervised NN label. Unsupervised selection does not use held-out test-label agreement.
  Do not award if: Labels are input to clustering/DR, held-out test labels determine unsupervised settings, or label use is unclear.

- **(2 pts) Downstream leakage controls.**
  Award if: The study demonstrates a correct train-only fitting protocol for learned preprocessing and DR/cluster features in Parts 4 and 5; validation/test data are transformed without refitting, and test outcomes do not guide model selection.
  Do not award if: The protocol uses full-sample fitted transforms for supervised testing, tunes with test results, or does not establish how leakage is prevented. This criterion addresses validity of the overall split protocol; avoid penalizing the identical mistake again under Part 5 unless a separate requirement also fails.

- **(2 pts) Testable hypothesis.**
  Award if: At least one hypothesis appears before the main results, predicts a meaningful method/data trade-off, and connects to theory or dataset properties.
  Do not award if: The hypothesis is absent, generic, purely a winner prediction without reasoning, or introduced only after seeing outcomes.

3 Part 1: Clustering on Original Features (12 points)

Goal: Assess K-Means and EM/GMM in original feature space and characterize what they capture.

- **(3 pts) K-Means model selection.**
  Award if: K-Means is applied to original BRFSS features with justified preprocessing, multiple reasonable k values, initialization/restart and seed settings, and a defensible selected value or comparison range.
  Do not award if: K-Means is missing; one arbitrary k is used; $k = 5$ is assumed correct because of the five labels; or important setup/model-selection evidence is absent.

- **(3 pts) EM/GMM model selection.**
  Award if: GMM is applied to original features with multiple reasonable component counts, an identified covariance model, a defensible selection rationale, and sufficient fitting/convergence details in the report or REPRO to reproduce the experiment.
  Do not award if: GMM is missing, component/model selection is arbitrary or unsupported, covariance type is unidentified, or essential fit information is insufficient for reproduction. Do not withhold all three points solely because a minor setting appears only in REPRO.

- **(2 pts) Internal clustering diagnostics.**
  Award if: At least two appropriate unlabeled/internal or model-selection diagnostics, such as silhouette, inertia/elbow, Davies–Bouldin, AIC, or BIC, are used to evaluate the clustering models and support interpretations.
  Do not award if: Metrics are absent, inappropriate, unrelated to choices, or only listed without meaningful interpretation.

- **(2 pts) Post-fit label alignment.**
  Award if: A post-fit quantitative or tabular comparison with `GenHlth` is provided (for example ARI, NMI, V-measure, or a contingency table), with at least a correct brief explanation of the result.
  Do not award if: No label-alignment evidence is supplied, the interpretation is misleading, or labels are used to tune the clustering fit. A short but accurate explanation still earns full points; suggest deeper analysis in feedback rather than denying the criterion.

- **(2 pts) Cluster characterization.**
  Award if: The report describes cluster sizes or profiles and at least one meaningful feature-related pattern, and explains why groups may align with or differ from General Health categories.
  Do not award if: The report gives only scores or cluster counts with no grounded explanation of what clusters represent.

4 Part 2: Dimensionality Reduction (14 points)

Goal: Explain representations created by PCA, ICA, and RP and justify selected dimensions.

- **(3 pts) PCA diagnostics.**
  Award if: PCA reports eigenvalues or explained variance, cumulative explained variance, a justified component count, and interpretation of preserved variance, redundancy, or feature structure.
  Do not award if: PCA is missing, component count is unsupported, explained-variance evidence is absent, or retained variance is equated with class information without support.

- **(3 pts) ICA diagnostics.**
  Award if: ICA reports a justified component count, kurtosis or another non-Gaussianity diagnostic, and relevant convergence/stability information with interpretation.
  Do not award if: ICA is missing, lacks non-Gaussianity evidence, hides convergence problems, or gives no component-count rationale.

- **(3 pts) RP diagnostics.**
  Award if: RP includes a justified projected dimension and distance preservation, distortion, or defensible reconstruction evidence, with variability across multiple seeds.
  Do not award if: RP is missing, is evaluated using only one unqualified favorable seed, or provides no diagnostic of preserved/distorted structure.

- **(3 pts) Cross-method dimension-selection rationale.**
  Award if: The report compares the selected PCA, ICA, and RP dimensions and explains why their compression levels or choices are comparable or intentionally different, using evidence from the explored settings and resulting trade-offs.
  Do not award if: There is no cross-method rationale for the selected dimensions, or the comparison is based on held-out test cherry-picking. Method-specific PCA/ICA/RP diagnostics are scored in their own items; do not deduct again here solely for a missing diagnostic already scored there.

- **(2 pts) Representation comparison.**
  Award if: A legible compact figure/table or integrated analysis compares what PCA, ICA, and RP preserve or lose, including the implications for later clustering or NN use.
  Do not award if: The methods are reported in isolation with no comparative representation analysis, or only downstream scores are given.

5 Part 3: Clustering after DR (12 points)

Goal: Evaluate how lower-dimensional representations change clustering behavior and external alignment.

- **(3 pts) All six combinations.**
  Award if: The report includes K-Means and GMM after each of PCA, ICA, and RP; results are labeled and verifiable for all six combinations.
  Do not award if: Any required pairing is absent, methods are unclear, or results cannot be associated with their pairing.

- **(2 pts) Post-DR cluster choices.**
  Award if: Cluster/component counts and relevant GMM settings are reselected or explicitly defended in each transformed space.
  Do not award if: Original settings are copied blindly across all reduced spaces without rationale.

- **(2 pts) Internal comparison evidence.**
  Award if: A compact, intelligible comparison of suitable clustering diagnostics connects the six reduced-space results with the original-space baselines.
  Do not award if: Internal results or baseline comparison are missing, fragmented, or uninterpretable.

- **(2 pts) External alignment comparison.**
  Award if: Post-fit `GenHlth` alignment is compared across original and reduced spaces using a metric or tabular evidence, with an appropriate interpretation.
  Do not award if: Alignment changes are asserted without evidence or not considered at all.

- **(2 pts) Representation-effect interpretation.**
  Award if: The report explains how PCA, ICA, or RP may have affected cluster quality, geometry, stability, or health-category agreement using findings from Parts 1 and 2.
  Do not award if: The analysis merely ranks six scores or offers no explanation tying representation to clustering.

- **(1 pt) Stability or comparison limitation.**
  Award if: At least one grounded limitation or variability issue is acknowledged, including seed effects, dimensionality, scaling, covariance assumptions, or non-comparable metrics across spaces.
  Do not award if: No relevant limitation or stability consideration is raised.

6 Part 4: Neural Network after DR (10 points)

Goal: Isolate the influence of dimensionality reduction on the established PyTorch NN.

- **(2 pts) Justified PyTorch NN recipe.**
  Award if: The model is based on the student's SL/OL PyTorch backbone and justified training recipe; input dimensions may change, while other training choices remain comparable or are explained.
  Do not award if: A different model family or a new unexplained architecture search replaces the SL/OL NN, or important training settings are missing.

- **(2 pts) Four required input conditions.**
  Award if: The same NN is evaluated on original, PCA, ICA, and RP inputs using the same data partitions and a fair comparison protocol.
  Do not award if: The original baseline or a required DR representation is missing, or conditions use unexplained incompatible splits or budgets.

- **(2 pts) NN evaluation metrics.**
  Award if: The original/PCA/ICA/RP comparison clearly reports held-out Accuracy, Macro-F1, balanced accuracy, and relevant validation-loss evidence under a consistent evaluation protocol.
  Do not award if: One or more required aggregate metrics or validation-loss evidence is missing, results cannot be matched to the four conditions, or the evaluation protocol is not credible.

- **(1 pt) Confusion matrix and class-level interpretation.**
  Award if: The main report includes an informative confusion matrix and discusses per-class precision, recall, and F1 in the five-class context. A full confusion matrix for every NN variant is not required if a compact comparison supports the conclusions.
  Do not award if: No confusion matrix is presented, or there is no substantive per-class precision/recall/F1 discussion.

- **(1 pt) Efficiency evidence.**
  Award if: Training time or another justified runtime/compute measure is compared between original and DR inputs, with preprocessing costs distinguished when material.
  Do not award if: No runtime/compute evidence is provided or the measure cannot support the stated comparison.

- **(2 pts) NN-after-DR interpretation.**
  Award if: The report explains the observed predictive and efficiency trade-offs in relation to compression, noise, lost information, or transformed geometry, without claiming unsupported superiority.
  Do not award if: The report merely lists outcomes, selects a winner without explanation, or ignores important class-level or cost changes.

7 Part 5: NN with Cluster-Derived Features (12 points)

Goal: Test whether original-space K-Means/GMM representations improve or change downstream NN learning.

- **(3 pts) Cluster-feature construction.**
  Award if: The report constructs both K-Means-derived and GMM-derived inputs from the original-space configurations investigated in Part 1, and clearly identifies feature types (e.g., one-hot assignment, centroid distance, posterior probability) and the chosen replacement or augmentation design.
  Do not award if: A clustering method is omitted, features are undefined, the student substitutes DR-space cluster models without justification, or the construction cannot be understood.

- **(2 pts) Part 1 configuration and feature mapping.**
  Award if: The K-Means and GMM settings are traceable to the raw-space selections in Part 1, and the construction/mapping of cluster-derived inputs across train, validation, and test observations is clear and reproducible.
  Do not award if: The feature-generating configurations are unrelated to Part 1 without explanation, or the mapping that produces NN features is unclear or unreproducible. Score train-only fitting and leakage under Study Design; do not remove these two points solely for the same leakage error if the Part 1 link and feature mapping are otherwise demonstrated.

- **(3 pts) Three comparable NN conditions.**
  Award if: Original-feature, K-Means-feature, and GMM-feature NNs are all evaluated using the same justified PyTorch recipe, sample splits, and comparable training budgets; necessary input-width changes are explained.
  Do not award if: Any required NN condition is missing, different network/training settings make the comparison uninterpretable, or the original NN baseline is absent.

- **(2 pts) Metrics and compute comparison.**
  Award if: The original/K-Means-feature/GMM-feature NNs are compared using Accuracy, Macro-F1, balanced accuracy, relevant validation-loss and class-level evidence, and runtime or compute. Class-level details may reuse a compact shared display rather than repeating full confusion matrices.
  Do not award if: The three-condition comparison is materially incomplete, relies only on Accuracy, lacks meaningful class-level discussion, or lacks a defensible compute comparison.

- **(2 pts) Cluster-feature interpretation.**
  Award if: The report explains whether cluster features add information, duplicate raw features, compress away predictive structure, or alter class-level errors, and connects findings with Part 1 clustering results.
  Do not award if: The report reports scores only or does not explain why the K-Means/GMM features helped, hurt, or had little effect.

8 Discussion (14 points)

Goal: Connect the five required parts in a scientific account of BRFSS feature structure and downstream utility. The Discussion should use prose and refer to quantitative evidence, not repeat a list of scores.

- **(2 pts) Original-space synthesis.**
  Award if: The report explains what unsupervised groups appear to represent in BRFSS based on cluster/feature evidence rather than simply ranking scores.
  Do not award if: The Discussion gives no meaningful conclusion about the original feature-space structure.

- **(2 pts) External-label synthesis.**
  Award if: The report discusses where clusters agree or disagree with `GenHlth` and what that says about unsupervised geometry versus health-category labels.
  Do not award if: Low or high ARI/NMI is stated without explanation, or cluster-label matching is treated as the sole definition of success.

- **(2 pts) K-Means versus GMM interpretation.**
  Award if: The report connects the different assumptions, hard versus soft membership, component shape, model selection, or observed profiles to meaningful findings.
  Do not award if: The methods are treated as interchangeable or only ranked numerically.

- **(2 pts) PCA, ICA, RP synthesis.**
  Award if: The report contrasts the three DR methods in terms of variance, non-Gaussianity, distance distortion, stability, component choice, and preservation of useful information as supported by its own evidence.
  Do not award if: There is no conceptual comparison of DR representations, or methods are ranked without explanation.

- **(2 pts) DR + clustering synthesis.**
  Award if: The report explains whether DR changed clustering quality or alignment and relates those changes to representation properties or limitations.
  Do not award if: The DR/clustering combinations are listed without a cross-part conclusion.

- **(1 pt) NN-after-DR synthesis.**
  Award if: The report explains whether DR improved, preserved, or harmed NN performance or efficiency in comparison with original features.
  Do not award if: The Discussion overlooks the supervised DR comparison or merely repeats scores.

- **(1 pt) NN cluster-feature synthesis.**
  Award if: The report explains whether K-Means/GMM feature construction adds or loses useful predictive structure and connects that conclusion to original-space clustering.
  Do not award if: Part 5 is not integrated into the Discussion or is summarized without meaning.

- **(1 pt) Specific limitations.**
  Award if: The report acknowledges evidence-based limitations such as sampling, class imbalance, scaling, Gaussian assumptions, instability, seed effects, or metric incompatibility.
  Do not award if: Limitations are absent or generic and not connected to findings.

- **(1 pt) Hypothesis resolution and takeaway.**
  Award if: The Discussion revisits the original hypothesis with evidence and ends with a defensible practical conclusion about what UL/DR/cluster features were useful for.
  Do not award if: The hypothesis is not resolved or the conclusion states only which final score was greatest.

Extra Credit: Nonlinear Manifold Learning (up to 5 points)

Students may use UMAP, t-SNE, Isomap, Laplacian Eigenmaps, or another justified nonlinear manifold method. Visualization-only use is acceptable. If used for downstream supervised NN inputs, the student must explain train/test-safe transformations.

- **(2 pts) Method implementation and rationale.**
  Award if: One nonlinear method is actually used, with a defensible choice and relevant parameter explanation; downstream use, if any, is leakage-safe.
  Do not award if: The method is only named or applied without a meaningful rationale, or downstream test information leaks into training.

- **(1 pt) Visualization or quantitative evidence.**
  Award if: A readable visualization or suitable quantitative diagnostic supports a claim about the nonlinear representation.
  Do not award if: No meaningful empirical evidence is provided.

- **(1 pt) Linear-method comparison.**
  Award if: The nonlinear outcome is compared with at least one required linear DR method.
  Do not award if: It is presented in isolation without context.

- **(1 pt) Interpretive insight.**
  Award if: The student explains whether or how the nonlinear method reveals structure not evident from the linear approaches.
  Do not award if: Interpretation is absent or limited to calling a plot visually interesting.

Version Control

- v1.1 – 11/09/2026 – TJL refactored UL rubric for Fall 2026.